1. PURPOSE:

1.1. To Cleaning & Sanitation of equipment's, Machine and under hygienic condition for Production.

1.2. This covers different Steps of CIP for Milk packing section, curd packing section, Curd processing section, paneer processing section.

2. SCOPE:

This procedure is applicable to West Assam Milk Producer's Cooperative Union Limited Guwahati.

3. RESPONSIBILITY:

3.1 Head Production & QA shall accountable for implementation of this procedure.

3.2 Deputy Manager/Sr. Executive/Executive shall responsible to maintain all activities as per required procedure.

3.3 Shift Chemist shall responsible to ensure the cleaning of lines, equipment and storage tanks.

4. PROCEDURE:

4.1 3 STEP HOT CIP"-For PMSTs & HMSTs

3 STEP HOT CIP-For PMSTs & HMSTs		
S. No.	**Step**	**Description**
1	Re-cup	Re-cup at 45-50 °C for 5 minutes
2	Hot Water	Sanitize with hot treated water at: • 90°C for 20 minutes OR • 85°C for 15 minutes
3	Soft Water Rinse	Final rinse with treated water till the equipment is cooled to below 35°C

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NOTE:

Intermittent Long Shut-Restart Procedure

In case of Post CIP, equipment is not used for more than 4 hours, then a 3 step CIP shall do before starting production.

Sl.No.	**Step**	**Water/CIP Solution**	**Supply/Return Temp.**	**Return Conductivity**		**Time**	**Total Time (In Minutes)**
				In%	**In Ms**	**Cycle Time (In Minutes)**	
1	Hot Water Sanitation	Hot Water	80-85°C (Return Temp)	NA	NA	10	<15
2	Final Rinse	Soft Water	Ambient Temp.	NA	NA	Till the equipment is cooled to below 35°c	5
						Total CIP Cycle Time	**20**

4.2 "5 Step CIP" SOP & Controls for heat treatment Equipment's (Silo's, PMST, HMST, Pasteurizers, Homogenizers):

Sl.No.	Step	Water/CIP Solution	Supply/Return Temp.	Return Conductivity		Time	Total Time (In Minutes)
				In%	In Ms	Cycle Time (In Minutes)	
1	Pre-Rinse	Re-cup.	45-50°C (Supply Temp.)	Na	Na	5	5
2	Lye Circulation	Hot Caustic	78-80°C (Return Temp)	1-1.2	55-70	30	<40
3	Fresh Water Rinse	Soft Water	30-35°C (Supply Temp.)	NA	NA	Conductivity Less Than 1ms+Additional 1 Minute For Stabilization	<10
4	Hot Water Sanitation	Hot Water	80-85°C (Return Temp)	NA	NA	20	<30
5	Final Rinse	Soft Water	Ambient Temp.	NA	NA	Till The Equipment Is Cooled To Below 35°C	5
						Total CIP Cycle Time	**90**

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Note:

Cycle time of CIP steps (i.e., step # 2, step # 4) should to be started only when the following conditions are met….

- Temperature at the coldest part of equipment i.e. in the return line
- Conductivity at the coldest part of equipment i.e. in the return line
- Flow requirement for the equipment to meet the required turbulence i.e. in the supply line
- Total time includes cycle time & time taken to achieve required temperature, conductivity, flow as well as delay times.

4.3 3.3. "7 Step CIP" SOP & Controls for heat treatment Equipment's (Silo's, PMST, HMST, Pasteurizers, Homogenizers:

Sl. No.	Step	Water/ CIP Solution	Supply/ Return Temp.	Return Conductivity		Time	Total Time (in minutes)
				In%	In Ms	cycle Time (in minutes)	
1	Pre-Rinse	Re-cup.	45-50°C (Supply Temp.)	NA	NA	5	5
2	Lye Circulation	Hot Caustic	75-80°C (Return Temp)	1-1.2	55-70	30	<40
3	Fresh Water Rinse	Soft Water	30-35°C(Supply Temp.)	NA	NA	Return Conductivity Less Than 1ms+Additional 1 Minute For Stabilization	<10
4	Acid Circulation	Hot Acid	70-75°C (Return Temp)	0.8-1.0	50-60	20	<30
5	Fresh Water Rinse	Soft Water	30-35°C (Supply Temp.)	NA	NA	Return Conductivity Less Than 1ms+Additional 1 Minute For Stabilization	<10
6	Hot Water Sanitation	Hot Water	80-85°C (Return Temp)	NA	NA	20	<30
7	Final Rinse	Soft Water	Ambient Temp.	NA	NA	Till The Equipment Is Cooled To Below 35°C	5
						Total CIP Cycle time	**130**

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Note:

Cycle time of CIP steps (i.e., step # 2, step # 4, step # 6) should to be started only when the following conditions are met….

- Temperature at the coldest part of equipment i.e. in the return line
- Conductivity at the coldest part of equipment i.e. in the return line
- Flow requirement for the equipment to meet the required turbulence i.e. in the supply line
- Total time includes cycle time & time taken to achieve required temperature, conductivity, flow as well as delay times.

4.4 Manual Cleaning SOP:

Manal Cleaning is required to remove milk solids/Dirt/Deposition, following steps Shall be followed.

Step	Description
Step-1	Proper rinsing with fresh warm water (45-50°C) to remove all loosely attached milk residues and other deposits if any.
Step-2	Add detergent (approved food grade formulated cleaning & sanitation chemical)- with suitable/recommended concentration in warm water and rub the surface using hygienic brush to loosen the residues.
Step-4	Rinsing with soft water to remove all chemical residues.
Step-4	Apply to 7 step CIP

NOTE:

Manual Cleaning shall always be performed before CIP, after manual cleaning, all the parts to be reassembled followed by CIP, as applicable.

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